

# Operating Systems

2520030042

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SEC-5

## WEEK-1

1. Develop a C program that demonstrates how a Linux operating system executes a command entered by a user that 1. Accept a Linux command as input. 2. Create a child process using `fork()`. 3. Execute the command in the child process using an appropriate `exec()` system call. 4. Allow the parent process to wait for the child using `wait ()`. 5. Display the Process ID (PID) of both parent and child processes.

Using Linux terminal commands (`uname`, `lscpu`, `lsblk`, `ps`, `top`), investigate the relationship between hardware resources and operating system services. Prepare a report explaining how the OS abstracts CPU, memory, storage, and I/O devices.

### Code

```
Ashish@Ashish: ~  
GNU nano 8.7.1  
#include <stdio.h>  
#include <unistd.h>  
#include <sys/wait.h>  
#include <stdlib.h>  
  
int main()  
{  
    char command[50];  
  
    printf("Enter command: ");  
    scanf("%s", command);  
  
    pid_t pid = fork();  
  
    if (pid == 0)  
    {  
        execlp(command, command, NULL);  
        perror("Command execution failed");  
        exit(1);  
    }  
    else if (pid > 0)  
    {  
        wait(NULL);  
        printf("Child process completed.\n");  
    }  
    else  
    {  
        perror("Fork failed");  
    }  
  
    return 0;  
}
```

### output

```
Ashish@Ashish:~$ nano linux.c
Ashish@Ashish:~$ gcc linux.c
Ashish@Ashish:~$ ./a.out
Enter command: ls
a.out  fork.c  linux.c  process.c.save
Child process completed.
```